SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: Artificial Intelligence **COURSE CODE: CSA17**

COURSE OUTCOME COVERED

CO2: *Experiment search and game-solving techniques such as Greedy Search, A*, CSP, Minimax, and Alpha-Beta pruning with heuristic functions for solving complex AI problems efficiently. (BL3)*

Assessment Tool Weightage: 35%

Duration: 1 Hour

QUESTIONS: 5 **Total Marks:50**

Annexure A

Scenario Based Assignment

Code of Conduct:

I Darani E (192312120) *certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:
DARANI E

Annexure A

Scenario Based Assignment

1. Scenario : A government deploys AI-powered drones to deliver emergency medicines in a flood-affected region. The environment is highly dynamic—roads may suddenly become inaccessible, and weather conditions affect travel cost. The drone has access to: A heuristic estimate (straight-line distance), Partial real-time updates about blocked paths and Variable movement costs depending on terrain and weather. However, due to urgency, the drone must quickly decide routes with minimum delay and risk.

Tasks:

i. Explain how Greedy Best-First Search would behave in this scenario and discuss its strengths and weaknesses under uncertainty

ii. Explain how A* would handle the same situation, including how it balances cost and heuristic

iii. Analyze the impact of heuristic accuracy on both algorithms

iv. Discuss how dynamic changes (blocked paths) affect search performance

v. Recommend the most suitable algorithm and justify your decision considering real-time constraints, optimality, and safety

ANSWER :

Greedy Best-First Search (GBFS) selects the path that appears closest to the destination based on a heuristic function.

- The drone uses straight-line distance to estimate how close each location is to the medicine delivery point.
- It selects the next location with the smallest heuristic value.
- It does not consider the total distance or cost already travelled.

Strengths:

- Fast decision-making.
- Requires less computational effort.
- Useful when quick route selection is necessary.

Weaknesses:

- May select a route with blocked roads.
- Ignores terrain and weather-related movement costs.
- Does not guarantee the shortest or safest route.
- May make poor decisions when real-time information is incomplete.

2. **Scenario**: A smart city implements an AI system to optimize traffic signal timings across hundreds of intersections. The objective is to minimize Total waiting time, Fuel consumption and Traffic congestion. The system must continuously adjust signals based on traffic flow. However the search space is extremely large, traffic patterns change dynamically and the system may get stuck in locally optimal solutions

Tasks. Formulate this as an optimization problem and explain why traditional search is inefficient

i. Explain how Hill Climbing would work in this scenario and identify its limitations

ii. Discuss issues like local maxima, plateaus, and ridges with real-world interpretation

iii. Explain how Simulated Annealing or Genetic Algorithms can overcome these limitations

iv. Recommend the best approach and justify your choice based on scalability and adaptability

ANSWER:

A* Search uses both the actual cost and the estimated cost to reach the destination.

It uses the formula:

$$f(n) = g(n) + h(n)$$

Where:

- $g(n)$ = Actual cost from the starting point to the current location.
- $h(n)$ = Estimated cost from the current location to the destination.
- $f(n)$ = Total estimated cost of the path.

Working:

1. The drone identifies possible routes.
2. It calculates the actual travel cost, including distance, weather and terrain.
3. It estimates the remaining distance using a heuristic.
4. It selects the route with the lowest $f(n)$.
5. It updates the route when new information becomes available.

A* provides a better balance between actual travel cost and estimated remaining cost.

3. **Scenario:** An autonomous rover is deployed on Mars to explore unknown terrain. It must: navigate safely to collect samples, avoid hazards (craters, rocks), operate with limited sensing capability and make decisions without a complete map. The rover updates its knowledge as it explores, making this a real-time decision-making problem.

Tasks:

i. Explain why this problem requires an online search agent instead of offline search

ii. Analyze the challenges of partial observability and dynamic environments

iii. Describe how the rover builds and updates its internal model

iv. Compare online search with uninformed and informed search strategies

v. Justify the most suitable approach considering uncertainty, adaptability, and safety

ANSWER:

Heuristic accuracy plays an important role in selecting efficient routes.

- In Greedy Best-First Search, an inaccurate heuristic may lead the drone towards a route that appears close but contains obstacles.
- In A*, an accurate heuristic improves route selection and reduces unnecessary exploration.
- An overestimated heuristic can cause A* to lose its optimality guarantee.
- An underestimated heuristic may increase the number of locations explored but can preserve optimality when it is admissible.

Therefore, accurate heuristic information improves search efficiency and route reliability.

4. **Scenario:** A university is designing an AI system to automatically generate exam timetables. The system must satisfy multiple constraints: No student should have overlapping exams, Limited number of exam halls, Certain subjects must be scheduled before others, Faculty availability constraints, Special accommodations for some students, The complexity increases as the number of courses and students grows.

Tasks :

Formulate the problem as a Constraint Satisfaction Problem (CSP)

i. Clearly define variables, domains, and constraints with explanation

ii. Explain how backtracking search works in this scenario

iii. Discuss how constraint propagation (e.g., forward checking) improves efficiency

iv. Analyze real-world challenges and justify the best strategy for scalability

ANSWER:

Dynamic changes can significantly affect both algorithms.

- Flooding may suddenly block previously available routes.
- Weather conditions may increase travel costs.
- The drone must receive updated information from sensors or communication systems.
- Previously selected routes may become unsuitable.
- Both algorithms may need to recalculate the route.

A* may require additional computation to find a new optimal route, while Greedy Best-First Search may quickly select another route but cannot guarantee its quality.

5. **Scenario:** Strategic Game AI for Real-Time Decision Making (Minimax & Alpha-Beta Pruning) A gaming company is developing an AI opponent for a real-time strategy game. The AI must complete against human players, Make decisions within strict time limits, Evaluate complex game states with multiple possible moves, The decision tree is extremely large, making computation expensive.

Tasks:

i. Explain how the Minimax algorithm models this problem

ii. Analyze the computational challenges of large game trees

iii. Explain how Alpha-Beta pruning improves efficiency with detailed reasoning

iv. Design an evaluation function and justify the factors considered

v. Recommend a strategy for balancing optimality and real-time performance

ANSWER:

Recommended Algorithm: A Search with Dynamic Route Replanning.*

A* is more suitable because:

- It considers both actual travel cost and estimated remaining distance.
- It can account for weather and terrain costs.
- It provides optimal solutions when the heuristic is admissible and the relevant cost information is accurate.
- It can recalculate routes when obstacles are detected.

For faster real-time operation, incremental algorithms such as D* Lite can be used to update routes when the environment changes.

Thus, A* with dynamic replanning provides a better balance between speed, optimality and safety for emergency medicine delivery.

RUBRICS FOR CALCULATION

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand